# Exploratory Data Analysis (EDA) Report

## Telecom Revenue Assurance – Synthetic Dataset

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## 1. Objective

The objective of this EDA was to **profile and validate the synthetic telecom dataset** before finalizing the leakage detection logic.

The analysis focused on:

- dataset scale and structure
- traffic distribution
- temporal consistency
- plan-wise behavior
- charge distribution
- data quality checks

This profiling phase helped validate whether the synthetic data supports realistic telecom revenue assurance use cases.

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## 2. Dataset Overview

The initial profiling of the CDR dataset produced the following summary:

| Metric | Value |
|---|---|
| Total CDRs | 49,525 |
| Unique Subscribers | 1,001 |
| Unique Plans | 6 |
| Service Types | 3 |

### Observation

The dataset contains **49.5k telecom usage records** across **1,001 subscribers** and **6 plans**, providing sufficient scale for rule-based leakage simulation.

The three service categories are:

- Voice
- Data
- SMS

This ensures balanced testing across multiple telecom usage types.

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# 3. Service-Type Distribution

| Call Type | Count |
|---|---|
| SMS | 16,612 |
| Data | 16,514 |
| Voice | 16,399 |

### Observation

The traffic is **almost evenly distributed** across all three service types.

This is a strong indicator of a well-balanced synthetic dataset and ensures that leakage detection logic is not biased toward any one usage category.

### Insight

Approximate distribution:

- SMS → 33.54%
- Data → 33.35%
- Voice → 33.11%

This near-uniform split is ideal for testing billing controls.

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# 4. Temporal Distribution Analysis

Daily CDR counts were analyzed for the period **1 Jan 2024 – 30 Jan 2024**.

The daily volume ranges from:

- **Minimum:** 1,588
- **Maximum:** 1,762
- **Average:** ~1,651

### Observation

The daily trend remains **highly stable**, with no missing dates and only minor day-to-day fluctuations.

This indicates:

- no data ingestion gaps
- no artificial spikes
- good temporal consistency

### Insight

The stability of daily traffic supports the reliability of downstream leakage trend analysis.

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## 5. Plan-Wise Usage Profiling

| Plan ID | Usage Count | Avg Duration | Avg Data MB |
|---|---|---|---|
| P2 | 9,913 | 100.56 | 8.53 |
| P3 | 9,818 | 99.92 | 8.32 |
| P4 | 9,801 | 99.35 | 8.20 |
| P1 | 9,780 | 105.27 | 8.08 |
| P0 | 9,705 | 95.70 | 8.65 |
| P5 | 508 | 102.38 | 8.04 |

### Observation

Plans **P0–P4** show an almost uniform traffic distribution.

Plan **P5** has significantly fewer records.

### Business Interpretation

P5 was intentionally used as a **special plan for the misclassification leakage use case (L9)**.

This validates the synthetic data design.

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## 6. Charge Distribution Analysis

| Metric | Value |
|---|---|
| Minimum Charge | 0.00 |
| Maximum Charge | 23.00 |
| Average Charge | 2.95 |
| Median Charge | 0.53 |

### Observation

The charge distribution is **strongly right-skewed**.

The large gap between average and median shows:

$$\text{Average Charge} > \text{Median Charge}$$

$$\text{Average Charge} > \text{Median Charge}$$

This suggests:

- large number of low-value events
- small number of high-charge outliers

### Business Interpretation

This closely resembles realistic telecom charging behavior.

It also strongly supports:

- **L2 → Wrong tariff**
- **L5 → Zero-rated anomaly**

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## 7. Data Quality Profiling

| Metric | Count |
|---|---|
| Null Duration | 511 |
| Null Start Time | 0 |
| Null End Time | 0 |

### Observation

A total of **511 records have missing duration values**, while timestamps remain fully populated.

### Business Interpretation

This directly supports the implementation of:

```
L12 → Partial CDR
```

This is a strong example of how EDA informed leakage rule design.

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# 8. Key Insights

The EDA revealed the following major findings:

1. The dataset is **well balanced** across service types
2. Daily traffic is **temporally stable**
3. Plan usage is **uniform except for the special L9 plan**
4. Charge distribution is **realistically skewed**
5. Missing duration values directly justify **partial CDR logic**

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# 9. Conclusion

The exploratory analysis phase helped validate the **structure, consistency, and realism of the synthetic telecom dataset**.

The findings from this EDA were directly used to support and justify the leakage detection rules implemented in the project.